# Land 4: Dermis Dunes

The integumentary system · OpenStax Ch. 5

## Key ideas

### Functions of the Skin

- Protection: a physical barrier of keratin, an acid mantle, dendritic immune cells and melanin against UV light.
- Prevents water loss through lipids released in the stratum granulosum.
- Thermoregulation through sweating and changes in dermal blood flow.
- Sensation through Merkel cells, Meissner and Pacinian corpuscles and free nerve endings.
- Starts vitamin D synthesis and excretes small amounts of salt and urea in sweat.

### Epidermis Layers

- Deep to superficial: stratum basale, spinosum, granulosum, lucidum, corneum.
- Stratum basale: one row of dividing stem cells plus melanocytes and Merkel cells.
- Stratum spinosum: spiny look from desmosomes; holds most dendritic (Langerhans) cells.
- Stratum granulosum: cells flatten, die and release lipid that waterproofs the skin.
- Stratum lucidum: thin clear layer found only in thick skin.
- Stratum corneum: 15 to 30 layers of dead keratinized cells that are constantly shed.

### Epidermal Cells and Skin Types

- Keratinocytes make keratin and are the most numerous epidermal cells.
- Melanocytes make melanin that caps keratinocyte nuclei to shield DNA from UV light.
- Merkel cells sense light touch; dendritic (Langerhans) cells present antigens to immune cells.
- Thick skin (palms and soles) has five strata and no hair; thin skin has four strata and hair follicles.
- The epidermis is avascular, so a cut that bleeds has reached the dermis.

### Dermis and Hypodermis

- Papillary layer: thin areolar tissue with dermal papillae, capillaries and Meissner corpuscles.
- Reticular layer: thick dense irregular connective tissue with collagen and elastic fibers for strength and recoil.
- Overstretching tears dermal fibers and leaves stretch marks (striae).
- Hypodermis: areolar and adipose tissue that anchors skin, stores fat, insulates and cushions.
- Subcutaneous injections go into the hypodermis and absorb more slowly than injections into muscle.

### Skin Color

- Melanin (brown to black and reddish types) is made by melanocytes; skin tone depends on melanin amount and type, not melanocyte number.
- Carotene from the diet gives a yellow orange tint.
- Hemoglobin in dermal blood gives a pink tone.
- Cyanosis is a bluish color from poorly oxygenated hemoglobin.
- Jaundice is yellowing of skin and sclera from bilirubin, often from liver dysfunction.
- Albinism: melanocytes are present but cannot make melanin.

### Hair, Nails and Glands

- Hair: shaft above the skin, root in the follicle, matrix in the bulb divides; the papilla supplies blood.
- Growth cycle: anagen (growth), catagen (regression), telogen (rest); arrector pili contraction causes goose bumps.
- Nails: new nail forms at the nail root (matrix); the nail bed's blood shows through as pink; eponychium is the cuticle; the lunula is the white crescent.
- Eccrine sweat glands are everywhere, release watery sweat through pores and cool the body.
- Apocrine glands in the axillae and anogenital region empty into hair follicles and become active at puberty.
- Sebaceous glands use holocrine secretion to release sebum into follicles; excess sebum contributes to acne.

### Vitamin D and Thermoregulation

- UV light converts a cholesterol derivative in the skin to vitamin D3 (cholecalciferol).
- The liver adds the next step and the kidney makes active calcitriol.
- Calcitriol increases calcium absorption from the intestine; deficiency weakens bones.
- When hot: dermal vessels dilate and eccrine sweat evaporates; high humidity slows evaporation.
- When cold: dermal vessels constrict to keep heat in the core.

### Wound Healing and Burns

- Shallow wounds that spare the stratum basale regenerate without scars.
- Deep wounds: clot, inflammation, granulation tissue, then scar (fibrosis).
- First degree: epidermis only, red and painful. Second degree: into the dermis, blisters and severe pain.
- Third degree: full thickness, may be painless at the center because nerve endings are destroyed.
- Rule of nines (adult): head 9, each arm 9, each leg 18, front trunk 18, back trunk 18, perineum 1 percent.
- Early danger after large burns is fluid and electrolyte loss leading to shock; infection follows.

### Skin Cancer, Pressure Injuries and Aging

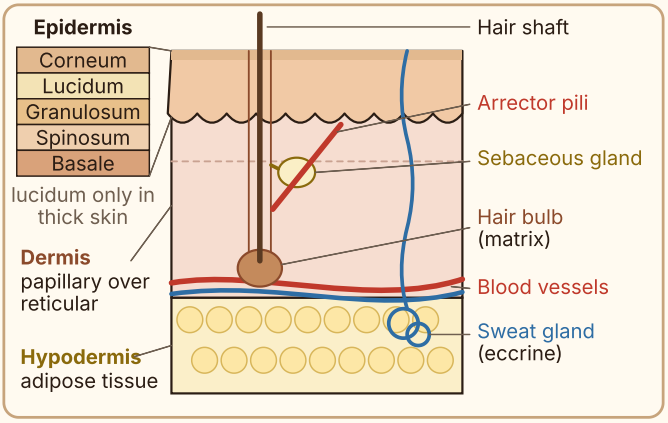
- Basal cell carcinoma: most common, from the stratum basale, pearly bump, rarely metastasizes.
- Squamous cell carcinoma: from stratum spinosum keratinocytes, can metastasize if untreated.
- Melanoma: from melanocytes, the deadliest; ABCDE is asymmetry, border, color, diameter over about 6 mm, evolving.
- Pressure injuries form over bony sites such as the sacrum and heels when sustained pressure blocks blood flow; turn patients every two hours.
- Aging skin: thinner dermis, less collagen and elastin, fewer active sweat glands, slower healing and easy bruising.

## Villager wisdom

- Old Mira says the stratum basale is the skin's nursery: its stem cells divide and push new keratinocytes up toward the surface, where they die and flake away.
- From the surface down, remember Come, Let's Get Sun Burned: corneum, lucidum, granulosum, spinosum, basale. The lucidum only appears in the thick skin of palms and soles.
- When the desert sun beats down, your dermal vessels widen and your eccrine glands pour out sweat. Evaporation carries the heat away from your body.
- Sunlight on the skin starts vitamin D, the liver adds the next piece, and the kidney finishes it as calcitriol so your gut can absorb calcium.
- For burns, each whole arm is 9 percent, each whole leg 18, the front of the trunk 18, the back 18, the head 9, and the perineum 1.

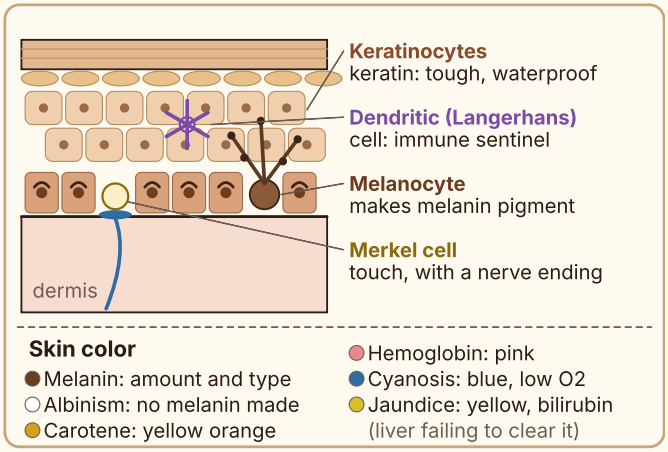
## Diagrams

### Layers of the skin



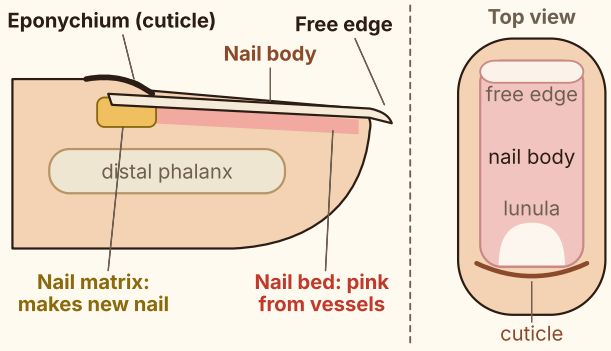

The epidermis is stratified squamous epithelium renewed from the stratum basale; the dermis holds glands, follicles and vessels; the fatty hypodermis insulates and cushions.

### Epidermal cells and skin color



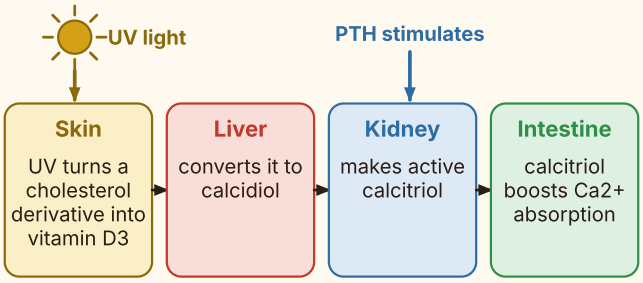

Keratinocytes make keratin, melanocytes make melanin, dendritic cells guard against microbes and Merkel cells sense touch; skin tone depends on the amount and type of melanin, not the number of melanocytes.

## Structure of a nail




New nail cells are made in the nail matrix at the root; the nail body looks pink because blood vessels in the nail bed show through, and the eponychium (cuticle) covers its proximal edge.

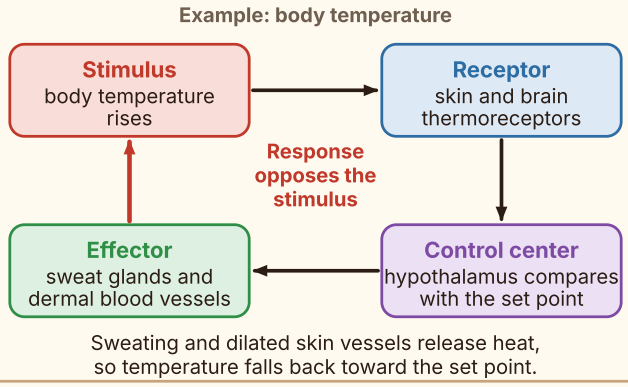
## Vitamin D synthesis




Too little UV (a housebound adult): low vitamin D, poor Ca2+ absorption and weak, soft bones

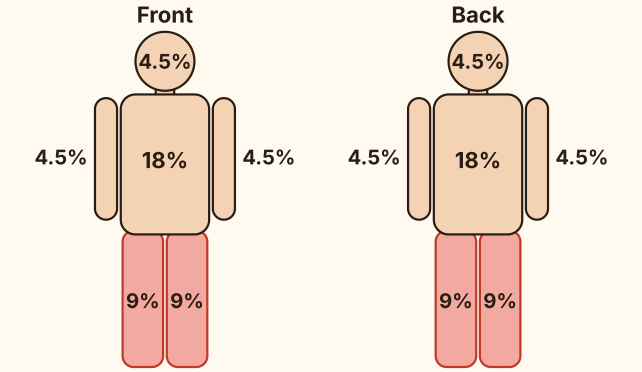
UV light starts vitamin D synthesis in the skin, the liver modifies it, and the kidney makes active calcitriol, which lets the intestine absorb calcium for strong bones.

## Negative feedback loop



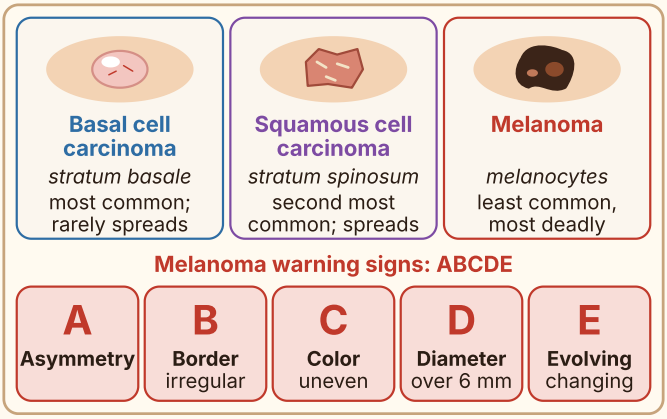

In negative feedback the effector response opposes the original change, returning the variable toward its set point; most homeostatic loops work this way.

## Rule of nines



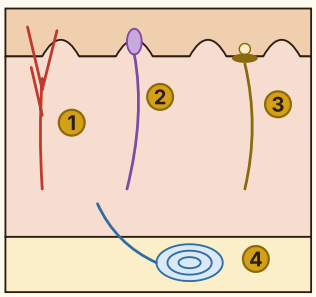

The rule of nines splits adult body surface into 9 percent blocks; both whole legs, front and back, total 36 percent.

## Skin cancers




Basal cell carcinoma is the most common and rarely spreads; melanoma, from melanocytes, is the most dangerous and is spotted with ABCDE: asymmetry, border, color, diameter over 6 mm, evolving.

## Sensory receptor types




1 Free endings: pain, heat
2 Meissner: light touch
3 Merkel disc: fine touch
4 Pacinian: deep pressure, vibration

**By stimulus**

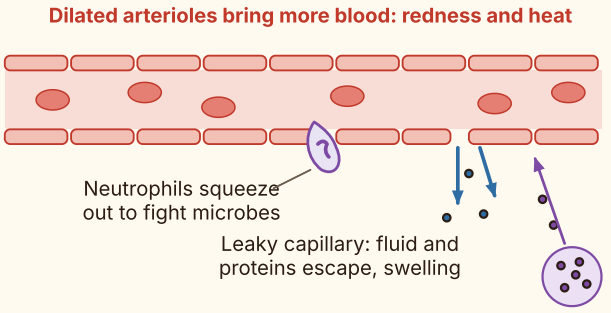
**Mechanoreceptors**
pressure, vibration, stretch
**Thermoreceptors**
temperature change
**Nociceptors**
tissue damage: pain
**Chemoreceptors**
chemicals: taste, smell
**Photoreceptors**
light: rods and cones

**By location**

**Exteroceptors**
outside stimuli, skin
**Interoceptors**
organs: bladder stretch
**Proprioceptors**
position: muscle spindles

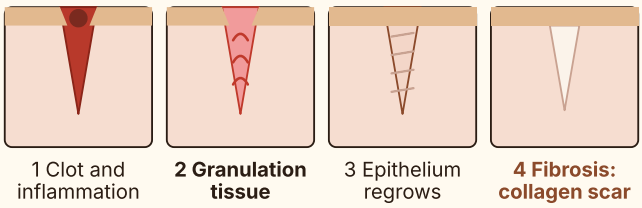
Receptors are classed by stimulus (mechanoreceptors, thermoreceptors, nociceptors, chemoreceptors, photoreceptors) and by location (exteroceptors, interoceptors, proprioceptors such as muscle spindles).

## Inflammation




Histamine dilates arterioles (redness, heat) and makes capillaries leaky, so fluid and proteins escape (swelling) and neutrophils move into the tissue.

## Tissue repair

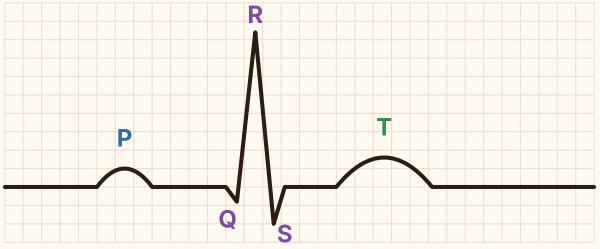



**Regenerate well**
epithelium, bone, loose connective tissue

**Mostly replaced by scar**
cardiac muscle and nervous tissue

A deep wound fills with a clot, then vascular granulation tissue; fibroblasts lay down collagen that becomes a scar (fibrosis), the usual outcome where tissue such as cardiac muscle cannot regenerate.

## The ECG




**P wave: atria depolarize**
**QRS complex: ventricles depolarize**
**T wave: ventricles repolarize**
Atrial repolarization is hidden in the QRS.

The P wave is atrial depolarization, the QRS complex is ventricular depolarization (hiding atrial repolarization), and the T wave is ventricular repolarization.

## Self check

1. **A housebound older adult who rarely goes outdoors develops weak bones. What skin related cause is most likely?**
    A. Low UV exposure limits vitamin D synthesis
    B. Too much melanin in the dermis
    C. Overactive sebaceous glands
    D. Loss of eccrine sweat glands
2. **The dermis is divided into which two layers?**
    A. Basale and corneum
    B. Superficial and deep fascia
    C. Hypodermis and epidermis
    D. Papillary and reticular
3. **Which epidermal layer consists of about 15 to 30 layers of dead, flattened, keratinized cells?**
    A. Stratum granulosum
    B. Stratum spinosum
    C. Stratum lucidum
    D. Stratum corneum
4. **In the ABCDE rule for spotting melanoma, the letter D stands for:**
    A. Depth of the mole in the dermis
    B. Diameter larger than about 6 mm
    C. Dark color that never changes
    D. Dryness of the surrounding skin
5. **Deep pressure and vibration are sensed by layered receptors in the deep dermis and hypodermis called:**
    A. Pacinian corpuscles
    B. Meissner corpuscles
    C. Merkel cells
    D. Free nerve endings
6. **Which sequence best describes healing of a deep skin wound?**
    A. Clot, inflammation, granulation tissue, scar
    B. Scar, clot, inflammation, granulation tissue
    C. Granulation tissue, clot, scar, inflammation
    D. Inflammation, scar, clot, granulation tissue
7. **On a hot day, a hiker has dilated dermal blood vessels and heavy sweating. What is the body accomplishing?**
    A. Increasing heat production
    B. Raising blood pressure
    C. Conserving body heat
    D. Releasing heat to lower body temperature

8. **Where is new nail tissue produced?**
    A. Nail body
    B. Hyponychium
    C. Nail root (matrix)
    D. Eponychium
9. **Which epidermal cells act as touch receptors linked to sensory nerve endings?**
    A. Dendritic (Langerhans) cells
    B. Melanocytes
    C. Merkel cells
    D. Keratinocytes
10. **What is the main function of sebum produced by sebaceous glands?**
    A. Cool the body by evaporation
    B. Absorb ultraviolet radiation
    C. Lubricate and waterproof skin and hair
    D. Carry wastes such as urea out
11. **A child with albinism has a normal number of melanocytes but almost no skin pigment. The problem most likely involves:**
    A. The melanocytes' ability to make melanin
    B. A total absence of melanocytes
    C. Excess carotene in the skin
    D. Too few dendritic cells
12. **When the body is cold, blood vessels in the dermis:**
    A. Dilate to release heat
    B. Leak plasma to warm the skin
    C. Constrict to reduce heat loss
    D. Grow new branches within minutes

## Answer key

1. **A.** Low UV exposure limits vitamin D synthesis. Without ultraviolet light, the skin makes little vitamin D, so less calcitriol is formed and the intestine absorbs less calcium. Melanin does reduce vitamin D production, but the stem points to lack of sun as the cause.
2. **D.** Papillary and reticular. The thin papillary layer of areolar tissue lies just under the epidermis, and the thick reticular layer of dense irregular connective tissue lies below it. The hypodermis is beneath the dermis and is not part of it.
3. **D.** Stratum corneum. The stratum corneum is the surface layer of dead keratinized cells that are continually shed. The stratum lucidum also has dead cells, but it is a thin clear layer found only in thick skin.
4. **B.** Diameter larger than about 6 mm. D is diameter, and a mole wider than about 6 mm, the size of a pencil eraser, is a warning sign. Dark color is covered by C for color, which looks for uneven shades rather than darkness alone.
5. **A.** Pacinian corpuscles. Pacinian (lamellated) corpuscles have onion like layers and respond to deep pressure and vibration. Free nerve endings are mostly receptors for pain and temperature.

6. **A.** Clot, inflammation, granulation tissue, scar. A clot first seals the wound, inflammation clears debris, granulation tissue fills the gap and then matures into scar. Scar formation is the final stage, so any order that starts with it is wrong.
7. **D.** Releasing heat to lower body temperature. Vasodilation brings warm blood near the surface to radiate heat, and evaporating sweat removes even more; this negative feedback lowers core temperature. Conserving heat would require vasoconstriction and reduced sweating instead.
8. **C.** Nail root (matrix). Dividing cells at the nail root form the nail body, which slides forward over the nail bed. The eponychium, or cuticle, is a fold of epidermis covering the root; it protects the root but does not make nail.
9. **C.** Merkel cells. Merkel cells in the stratum basale contact sensory nerve endings and respond to light touch. Dendritic cells are immune cells that capture antigens rather than sense touch.
10. **C.** Lubricate and waterproof skin and hair. Sebum is an oily mixture that keeps skin and hair soft, limits water loss, and has some antibacterial action. Melanin, not sebum, is the pigment that absorbs ultraviolet radiation.
11. **A.** The melanocytes' ability to make melanin. In albinism, melanocytes are present but a genetic defect, usually in the enzyme tyrosinase, prevents melanin synthesis. The stem rules out a lack of melanocytes because their number is normal.
12. **C.** Constrict to reduce heat loss. Vasoconstriction in the dermis keeps warm blood in the body core, so less heat is lost at the surface. Dilation is the response to overheating, when the body needs to shed heat.

Kingdom of Corpus · Dr. Akash Garg, Dr. Garg Learning · Aligned to OpenStax Anatomy and Physiology 2e · CC BY SA 4.0